Yun-Hung (Victor) Gao

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BIOGRAPHY

I was born in Hong Kong SAR, China, in 2002. I will join **The University of Texas at Austin (UT Austin)** as a Postdoctoral Fellow in December 2026. I received dual bachelor's degrees from **Peking University (PKU)** and earned my Ph.D. from **The Chinese University of Hong Kong (CUHK)** with early graduation in 3 years at the age of 24.

RESEARCH INTERESTS

- Power-efficient mixed-signal IC and SoC design for biomedical and optical applications
- Direct-digitization based versatile sensor interface and non-idealities cancellation techniques
- Software-hardware co-designs for on edge AI computing and acceleration

POSTDOCTORAL POSITION

The University of Texas at Austin Austin, United States
Incoming Postdoctoral Fellow *(Department of Electrical and Computer Engineering)* Dec. 2026 – Dec. 2027 **(Contract)**

- Advisor: Prof. Yaoyao Jia

EDUCATION

The Chinese University of Hong Kong Hong Kong SAR, China
Ph.D. in Electronic Engineering *(Department of Electronic Engineering)* Aug. 2023 – **Oct. 2026 (Early Graduation)**

- Advisor: Prof. Kong-Pang Pun
- Thesis title: Successive-approximation-register based light-to-digital converter design for biomedical applications

Peking University Beijing, China
B.E. in IC Design and Integrated System *(School of Electronic Engineering and Computer Science)* Sep. 2019 – Jul. 2023
B.M. in Management of Innovation and Entrepreneurship *(Guanghua School of Management)* Sep. 2020 – Jul. 2023

PROFESSIONAL EXPERIENCE

The Chinese University of Hong Kong Hong Kong SAR, China
Research Assistant *(Department of Electronic Engineering)* Oct. 2026 – Present

Hong Kong Centre for Cerebro-Cardiovascular Health Engineering Hong Kong SAR, China
Student Intern Jul. 2022 – Aug. 2022

Shanghai IC R&D Center Shanghai, China
Analog IC Design Intern Jan. 2022 – Mar. 2022

Dekean Electronics (Ambiq Micro) Shanghai, China
IC Verification Intern July. 2021 – Sep. 2021

SELECTED AWARDS AND HONORS

- **Highlighted Paper of August 2026**, *IEEE Transactions on Circuits and Systems I: Regular Papers* 2026
- **CUHK EE Department Postgraduate Student Research Excellence Award (3/~160; first student from the VLSI and ASIC group to receive this award)** 2026
- For Our Future Innovation and Technology Scholarship **(1 of 10 awardees in Hong Kong; sole awardee from the CUHK School of Engineering)** 2025
- CUHK EE Department Tutor Commendation Award 2024
- **Anantha Chandrakasan Award for Distinguished-Technical Paper,** 2023 *IEEE International Solid-State Circuits Conference (ISSCC)* **(co-recipient; first team from China to receive this award)** 2024
- Peking University Grand Prize Scholarship for the Hong Kong, Macau, and Overseas Chinese Undergraduate Student **(1/~260; sole awardee of the year)** 2022
- Hong Kong Centre for Cerebro-Cardiovascular Health Engineering Young Scientist Grant Award 2022
- Peking University Shu Qi Scholarship 2022
- Peking University Research Excellence Award 2022

PUBLICATIONS

Journal and Conference Papers:

[1] **Y. -H. Gao**, C. -H. Fan, J. Li, S. Li, Z. Jin, K. N. Leung, Y. -T. Zhang X. Tang and K. -P. Pun, "A noise-shaping-SAR multiplexed light-to-digital converter with passive nested feedback and predictive baseline current compensation achieving 95-dB SNDR and 161.5-dB DR," ***IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)***, vol. 73, no. 8, pp. 5168-5181, Aug. 2026 **[2026 IEEE TCAS-I Highlighted Papers]**

[2] **Y. -H. Gao**, J. Li, C. -H. Fan, K. N. Leung, Y. -T. Zhang and K. -P. Pun, "A 9.84-μW 148.9-dB total DR light-to-digital converter with current-integration SAR quantizer for multi-wavelength PPG applications," ***IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)***, vol. 72, no. 11, pp. 6886-6899, Nov. 2025. **[2025 CUHK EE Postgraduate Student Research Excellence Award]**

[3] **Y. -H. Gao**, Y. -T. Zhang, Q. Wen, J. Li and K. -P. Pun, "A 90.9-dB SNDR fourth-order fully passive bandpass current-integrating noise-shaping SAR light-to-digital converter," ***IEEE Transactions on Very Large Scale Integration Systems (TVLSI)***, vol. 34, no. 9, pp. 3060-3064, Sept. 2026.

[4] **Y. -H. Gao** et al., "A bit-level-sparsity-aware noise-shaping SAR based PPG sensor front-end for efficient backend multiply-accumulate interfaces", ***IEEE Sensors Journal***, accepted, DOI: 10.1109/JSEN.2026.3741037.

[5] **Y. -H. Gao**, C. -H. Fan, J. Li, K. N. Leung, Y. -T. Zhang and K. -P. Pun, "A reconfigurable LP/BP/HP current-sensing front-end achieving 109.3-dB SFDR and tolerance to error current and sensor parasitics with adaptive auto-zeroing scheme embedded in SC-CMFB," in ***IEEE Custom Integrated Circuits Conference (CICC)***, Seattle, WA, USA, 2026, pp. 1–4.

[6] **Y. -H. Gao**, H. Li, K. Zhang, X. Yu, H. Lu, "Mortar: morphing the bit level sparsity for general purpose deep learning acceleration", in ***IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)***, Tokyo, Japan, 2023, pp. 739-744.

[7] **Y. -H. Gao**, K. Zhang, S. Jia, "BEM: bit-level sparsity-aware deep learning accelerator with efficient booth encoding and weight multiplexing", in ***IEEE International Conference on Circuits and Systems***, Chengdu, China, 2022, pp. 187-194.

[8] J. Gao, L. Shen, H. Li, S. Ye, J. Li, X. Xu, J. Cui, **Y. -H. Gao**, R. Huang, L. Ye, "A 7.9fJ/ conversion-step and 37.12aFrms pipelined-SAR capacitance-to-digital converter with kT/C noise cancellation and incomplete-settling-based correlated level shifting", in ***IEEE International Solid-State Circuits Conference (ISSCC)***, San Francisco, CA, USA, 2023, pp. 346–348. **[2023 ISSCC Distinguished-Technical Paper Award]**

Working Papers:

[9] **Y. -H. Gao** et al., "A fully passive reconfigurable passband noise-shaping SAR based current-sensing front-end with hardware-reused adaptive auto-zeroing scheme," under preparation, ***IEEE Journal of Solid-State Circuits (JSSC).***

[10] **Y. -H. Gao** et al., "A second order passive nested feedback kT/C noise-shaping SAR light-to-digital converter with reconfigurable passband", under preparation," ***IEEE Transactions on Very Large Scale Integration Systems (TVLSI)***

[11] **Y. -H. Gao** et al., "WBAP: weight-bucketized-activation-pruning for bit-level sparsity-aware deep learning on-edge reconfiguration and acceleration," under preparation, ***IEEE Journal on Emerging and Selected Topics of Circuits and Systems (JETCAS)***

SELECTED PATENTS

[P1] **Y. -H. Gao**, S. Jia, and J. Duan, "Computing unit, array circuit and control method for in-memory computing based on memristors," China Patent, CN114662682B.

[P2] **Y. -H. Gao**, S. Jia, and J. Duan, "Timing-synchronous computing unit and method for in-memory computing based on memristors," China Patent, CN115019852B.

[P3] **Y. -H. Gao** and J. Duan, "In-memory computing unit and in-memory computing array based on memristors," China Patent, CN115019854B.

[P4] **Y. -H. Gao** and J. Duan, "Multi-bit in-memory computing unit and control method based on memristors," China Patent, CN115019853B.

[P5] Y. Zhang, T. Xiang, N. Ji and **Y. -H. Gao**, "Nail sensing based, wireless multi-modal tonoarteriogram monitoring apparatus and system", US Patent Application, US20240081738A1.

[P6] H. Yang, **Y. -H. Gao**, C. Li, and L. Guo, "In-memory computing signal processing circuit," China Patent Application, CN202210737192.9.

RESEARCH EXPERIENCE

School of Integrated Circuits @ Peking University Beijing, China

Research Intern *(ASIC Lab)* Sep. 2022 – Jul. 2023

- Advisor: Prof. Wengao Lu and Prof. Yacong Zhang

Research Intern *(AIOT Lab)* Mar. 2022 – Jul. 2022
- Advisor: Prof. Le Ye

Research Intern *(MPW Lab)* Feb. 2022 – Jul. 2023
- Advisor: Prof. Song Jia

State Key Laboratory of Computer Architectures @ Chinese Academy of Science Beijing, China
Research Intern *(HPU Lab)* Jan. 2022 – Sep. 2022
- Advisor: Prof. Hang Lu

RESEARCH HIGHLIGHTS

Current-Integration SAR Based Light-to-Digital Converter for Photoplethysmography Applications

First Work:
- Prototype fabricated in a 180-nm CMOS process.
- Key concept: Introduces the first CI-SAR based LDC reusing CDAC in SAR as the integrator's feedback and employing an AC/DC correlated DR enhancement loop.
- Achieves 148.9-dB DR while consuming 9.84-μW AFE power.
- Published in *IEEE Transactions on Circuits and Systems I: Regular Papers* 2025 [2] (first authored).

Second Work:
- Prototype fabricated in a 180-nm CMOS process.
- Key concept: Introduces a fully passive nested-feedback noise-shaping SAR (NS-SAR) based LDC with predictive digital baseline-compensation loop filter.
- Achieves a 95-dB SNDR, 112.4-dB SFDR and 161.5-dB total DR while consuming 26.5-μW AFE power.
- Achieves the state-of-the-art FoM_{DR} of 220.3-dB among reported DT LDCs.
- Published in *IEEE Transactions on Circuits and Systems I: Regular Papers* 2026 [1] (first authored).

Bit-level Sparsity Aware Software-Hardware Co-Design for Edge DNN Inference Applications

First Work:
- Prototype synthesized in a 28-nm CMOS process.
- Key concept: Introduces a mantissa morphing approach that exploits the bit-level sparsity within weights for DNN inference acceleration.
- Achieves 1.28–2.51× bit-level sparsity improvement on weights with −0.09–0.23% model accuracy loss.
- Achieves up to 4.8× inference speedup by accelerator with a 0.031 mm² area and 68.58 mW power.
- Presented in *IEEE/ACM Asia and South Pacific Design Automation Conference* 2023 [6] (first authored).

Second Work:
- Prototype synthesized in a 40-nm CMOS process.
- Key concept: Introduces a weight-bucketized activation-pruning approach that exploits the bit-level sparsity within activations for DNN inference acceleration.
- Achieves an average 2.25× bit-level sparsity improvement on activations with negligible model accuracy loss.
- Achieves up to 6.57× inference speedup by accelerator with a 0.29 mm² area and 15.88 mW power.
- Plan to submit to *IEEE Journal on Emerging and Selected Topics of Circuits and Systems* [11] (first authored).

Third Work:
- Prototype fabricated in a 180-nm CMOS process.
- Key concept: Introduces a bit-level sparsity-aware PPG sensor with efficient booth-encoding SAR logic.
- Achieves 144.7-dB DR over a 1-kHz bandwidth while consuming 42.4-μW.
- Achieves up to 61.4% savings in MAC operations with only sub-μW overhead at the LDC level.
- Published in *IEEE Sensors Journal 2026* [4] (first authored).

Current-to-Digital Converter for Versatile Current-Readout Applications

First Work:
- Prototype fabricated in a 180-nm CMOS process.
- Key concept: Introduces an LP/BP/HP CI NS-SAR IDC with a reconfigurable passive loop filter and an adaptive auto-zeroing scheme embedded in the integrator for input offset and sensor parasitic capacitance compensation.
- Achieves 94.3/81.5/93.4 dB-SNDR and 105.7/86.8/109.3-dB SFDR in LP/BP/HP modes over a 500-Hz BW, with AC DR of 106.1/104.2/103.1 dB and total DR of 148.8 dB in LP mode, while consuming 37.6–39.8 μW AFE power.

- Achieves up to ±6400% integrator input mismatch tolerance with <5 dB SNDR loss.
- Achieves 93.4-dB SNDR and 109.3-dB SFDR in HP mode under an intentional 400% integrator input mismatch.
- Presented in *IEEE Custom Integrated Circuits Conference* 2026 [5] (first authored).

Second Work:

- Prototype fabricated in a 180-nm CMOS process.
- Key concept: Introduces a 4th order BP CI NS-SAR IDC with a fully passive loop filter.
- Achieves 90.9-dB SNDR and 107.2-dB DR while consuming 35.41-μW AFE power.
- Published in *IEEE Transactions on Very Large Scale Integration (VLSI) Systems* 2026 [3] (first authored).

PROFESSIONAL SERVICE

Journal Reviews:

- IEEE Journal of Solid-State Circuits (JSSC)
- IEEE Transactions on Biomedical Circuits and Systems (TBioCAS)
- IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II)
- Nature Scientific Reports (SR)

TEACHING EXPERIENCE

- ELEG 5764 Artificial Intelligence IC Design
- ELEG 4211 CMOS Digital Integrated Circuits Design
- ELEG 3201 Microelectronic Devices and Circuits
- ELEG 2201 Digital Circuits and Computing Systems

REFERENCES

Prof. Kong-Pang Pun, The Chinese University of Hong Kong, Hong Kong SAR, China

- Relationship: Ph.D. advisor
- Email: kppun@ee.cuhk.hk

Prof. Ka Nang Leung, The Chinese University of Hong Kong, Hong Kong SAR, China

- Relationship: Academic collaborator
- Email: knleung@ee.cuhk.hk